

CBSE Class 10 Mathematics — Circles — Solutions

1. 1. A tangent PQ at a point P of a circle of radius 5 cm meets a line through the center O at a point Q so that $OQ = 13$ cm. The length of PQ is:

Answer: (C)

Solution:

1. The radius is perpendicular to the tangent at the point of contact, so $\angle OPQ = 90^\circ$.
2. In right-angled triangle $\triangle OPQ$, by Pythagoras theorem: $OQ^2 = OP^2 + PQ^2$.
3. Substitute $OP = 5$ and $OQ = 13$: $13^2 = 5^2 + PQ^2$.
4. Solve for PQ : $169 = 25 + PQ^2 \implies PQ^2 = 144 \implies PQ = 12$ cm.

Derivation:

$$PQ = \sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144} = 12$$

Circles — Tangents to a circle

2. 2. If tangents PA and PB from a point P to a circle with center O are inclined to each other at an angle of 80° , then $\angle POA$ is equal to:

Answer: (A)

Solution:

1. The line OP bisects the angle between the tangents, so $\angle APO = 40^\circ$.
2. In $\triangle OAP$, $\angle OAP = 90^\circ$ (radius is perpendicular to tangent).
3. In $\triangle OAP$, $\angle POA + \angle OAP + \angle APO = 180^\circ$.
4. $\angle POA + 90^\circ + 40^\circ = 180^\circ \implies \angle POA = 50^\circ$.

Derivation:

$$\angle POA = 180^\circ - 90^\circ - 40^\circ = 50^\circ$$

Circles — Tangents from an external point

3. 3. The number of tangents that can be drawn from a point lying inside a circle is:

Answer: (A)

Solution:

1. A point inside a circle does not lie on the circumference.
2. A line passing through an interior point will be a secant, not a tangent.
3. Therefore, zero tangents can be drawn.

Derivation: By definition, a tangent touches the circle at exactly one point on the circumference.

Circles — Tangent properties

4. **4.** Two concentric circles are of radii 5 cm and 3 cm. The length of the chord of the larger circle which touches the smaller circle is:

Answer: (B)

Solution:

1. Let O be the center, AB be the chord of the larger circle touching the smaller circle at M .
2. In $\triangle OMA$, $\angle OMA = 90^\circ$, $OA = 5$ cm, $OM = 3$ cm.
3. By Pythagoras: $AM^2 = OA^2 - OM^2 = 25 - 9 = 16$. So $AM = 4$ cm.
4. The perpendicular from the center bisects the chord, so $AB = 2 \times AM = 8$ cm.

Derivation:

$$AB = 2\sqrt{5^2 - 3^2} = 2\sqrt{16} = 8$$

Circles — Concentric circles

5. **5.** If TP and TQ are two tangents to a circle with center O so that $\angle POQ = 110^\circ$, then $\angle PTQ$ is equal to:

Answer: (C)

Solution:

1. In the quadrilateral $OPTQ$, the sum of angles is 360° .
2. Tangents are perpendicular to radii at contact points, so $\angle OPT = 90^\circ$ and $\angle OQT = 90^\circ$.
3. $\angle PTQ + \angle POQ + \angle OPT + \angle OQT = 360^\circ$.
4. $\angle PTQ + 110^\circ + 90^\circ + 90^\circ = 360^\circ \implies \angle PTQ = 360^\circ - 290^\circ = 70^\circ$.

Derivation:

$$\angle PTQ = 180^\circ - 110^\circ = 70^\circ$$

Circles — Tangents from an external point

6. **6.** Assertion (A): The length of a tangent drawn from an external point to a circle is always equal to the radius of the circle. Reason (R): Tangents drawn from an external point to a circle are equal in length.

Answer: (D)

Solution:

1. The length of the tangent from an external point is not necessarily equal to the radius.
2. The theorem states that tangents drawn from an external point to a circle are equal in length.
3. Therefore, Assertion is false and Reason is true.

Derivation: A: False, R: True

Circles — Tangents to a circle

7. **7.** Assertion (A): If the angle between two tangents drawn from a point P to a circle of radius r and center O is 60° , then $OP = 2r$. Reason (R): The line segment joining the center to the external point bisects the angle between the two tangents.

Answer: (A)

Solution:

1. In $\triangle OAP$ (where A is point of contact), $\angle OPA = 60^\circ/2 = 30^\circ$.
2. $\sin(30^\circ) = r/OP$, so $1/2 = r/OP$, which implies $OP = 2r$.
3. The reason explains the bisection property used in the calculation.

Derivation:

$$\begin{aligned}\sin(30^\circ) &= \frac{r}{OP} \\ \implies OP &= 2r\end{aligned}$$

Circles — Tangents to a circle

8. **8.** Assertion (A): A tangent to a circle is perpendicular to the radius at the point of contact. Reason (R): The perpendicular distance from the center of the circle to the tangent is equal to the radius of the circle.

Answer: (A)

Solution:

1. The theorem of tangents states that the tangent at any point of a circle is perpendicular to the radius through the point of contact.
2. The shortest distance from the center to the line (tangent) is the radius, which confirms the perpendicularity.
3. Both statements are true and R explains A.

Derivation: $\text{Radius} \perp \text{Tangent}$

Circles — Tangents to a circle

9. **9.** Assertion (A): The number of tangents that can be drawn from a point inside a circle is zero. Reason (R): A tangent to a circle intersects the circle at only one point.

Answer: (A)

Solution:

1. A point inside a circle cannot have a line touching the circle at only one point.
2. A tangent is defined as a line intersecting the circle at exactly one point.
3. Both are true and R explains why no tangents can exist from an internal point.

Derivation: $\text{Internal point} \implies 0 \text{ tangents}$

Circles — Tangents to a circle

10. **10.** Assertion (A): If the distance between two parallel tangents of a circle of radius 3 cm is 6 cm. Reason (R): Parallel tangents touch the circle at the ends of the diameter.

Answer: (A)

Solution:

1. The distance between two parallel tangents is equal to the diameter of the circle.

2. Diameter = $2 \times \text{radius} = 2 \times 3 = 6$ cm.
3. Both are true and R explains why the distance is the diameter.

Derivation:

$$d = 2r = 2(3) = 6$$

Circles — Tangents to a circle

11. 11. If a tangent PA and PB from a point P to a circle with center O are inclined to each other at an angle of 80° , then find $\angle POA$.

Answer:

$$50^\circ$$

Solution:

1. In quadrilateral $OAPB$, $\angle OAP = \angle OBP = 90^\circ$ because the radius is perpendicular to the tangent at the point of contact.
2. In $\triangle OAP$ and $\triangle OBP$, OP bisects $\angle APB$, so $\angle APO = 40^\circ$. In $\triangle OAP$, $\angle POA = 180^\circ - (90^\circ + 40^\circ) = 50^\circ$.

Derivation:

$$\begin{aligned} \angle APB &= 80^\circ \\ \implies \angle APO &= 40^\circ. \text{ In } \triangle OAP, \angle OAP = 90^\circ. \angle POA = 180^\circ - 90^\circ - 40^\circ = 50^\circ. \end{aligned}$$

Circles — Tangents to a circle

12. 12. A quadrilateral $ABCD$ is drawn to circumscribe a circle. If $AB = 6$ cm, $BC = 7$ cm, and $CD = 4$ cm, find the length of AD .

Answer:

$$3 \text{ cm}$$

Solution:

1. For a quadrilateral circumscribing a circle, the sums of opposite sides are equal: $AB + CD = BC + AD$.
2. Substitute the given values: $6 + 4 = 7 + AD$. Solving for AD gives $10 - 7 = 3$.

Derivation:

$$\begin{aligned} AB + CD &= BC + AD \\ \implies 6 + 4 &= 7 + AD \\ \implies AD &= 10 - 7 = 3 \text{ cm.} \end{aligned}$$

Circles — Properties of tangents

13. 13. From a point Q , the length of the tangent to a circle is 24 cm and the distance of Q from the center is 25 cm. Find the radius of the circle.

Answer:

$$7 \text{ cm}$$

Solution:

1. Let O be the center and A be the point of contact. $\triangle OAQ$ is a right-angled triangle with $\angle OAQ = 90^\circ$.
2. By Pythagoras theorem, $OQ^2 = OA^2 + AQ^2$. Substitute $25^2 = r^2 + 24^2$ and solve for r .

Derivation:

$$\begin{aligned}
 25^2 &= r^2 + 24^2 \\
 \implies 625 &= r^2 + 576 \\
 \implies r^2 &= 49 \\
 \implies r &= 7 \text{ cm.}
 \end{aligned}$$

Circles — Tangents to a circle

- 14. 14.** Two concentric circles are of radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle.

Answer:

$$8 \text{ cm}$$

Solution:

1. Let AB be the chord of the larger circle touching the smaller circle at M . $OM \perp AB$ and $OM = 3$ cm, $OA = 5$ cm.
2. In right $\triangle OMA$, $AM = \sqrt{OA^2 - OM^2} = \sqrt{25 - 9} = 4$. Since M is the mid-point, $AB = 2 \times 4 = 8$.

Derivation:

$$\begin{aligned}
 AM &= \sqrt{5^2 - 3^2} = \sqrt{16} = 4 \\
 \implies AB &= 2 \times 4 = 8 \text{ cm.}
 \end{aligned}$$

Circles — Tangents to a circle

- 15. 15.** If the angle between two tangents drawn from an external point P to a circle of radius r and center O is 60° , find the length of OP .

Answer:

$$2r$$

Solution:

1. Let the tangents be PA and PB . OP bisects $\angle APB$, so $\angle APO = 30^\circ$.
2. In $\triangle OAP$, $\sin(30^\circ) = \frac{OA}{OP} = \frac{r}{OP}$. Since $\sin(30^\circ) = 1/2$, we get $OP = 2r$.

Derivation:

$$\begin{aligned}
 \sin(30^\circ) &= \frac{r}{OP} \\
 \implies \frac{1}{2} &= \frac{r}{OP} \\
 \implies OP &= 2r.
 \end{aligned}$$

Circles — Tangents to a circle

16. **16.** Two concentric circles are of radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle.

Answer:

8 cm

Solution:

1. Let O be the center, AB be the chord of the larger circle tangent to the smaller circle at point P.
2. OP is the radius of the smaller circle, so $OP = 3$ cm and $OP \perp AB$.
3. OA is the radius of the larger circle, so $OA = 5$ cm.
4. In right triangle OPA, $AP^2 = OA^2 - OP^2 = 5^2 - 3^2 = 16$, so $AP = 4$ cm.
5. Since the perpendicular from the center bisects the chord, $AB = 2 \times AP = 8$ cm.

Derivation:

$$AP = \sqrt{OA^2 - OP^2} = \sqrt{25 - 9} = \sqrt{16} = 4 \text{ cm}; AB = 2 \times 4 = 8 \text{ cm}$$

Circles — Tangent properties

17. **17.** A quadrilateral ABCD is drawn to circumscribe a circle. Prove that $AB + CD = AD + BC$.

Answer:

$$AB + CD = AD + BC$$

Solution:

1. Use the property that lengths of tangents drawn from an external point to a circle are equal.
2. Let the circle touch sides AB, BC, CD, DA at P, Q, R, S respectively.
3. $AP = AS, BP = BQ, CR = CQ, DR = DS$.
4. Summing the left and right sides of these equations leads to the result.

Derivation:

$$AB + CD = (AP + PB) + (CR + RD) = AS + BQ + CQ + DS = (AS + DS) + (BQ + CQ) = AD + BC$$

Circles — Tangents to a circle

18. **18.** If a tangent PA and PB from a point P to a circle with centre O are inclined to each other at an angle of 80° , find $\angle POA$.

Answer:

50°

Solution:

1. In quadrilateral OAPB, $\angle OAP = \angle OBP = 90^\circ$ (tangent is perpendicular to radius).
2. Sum of angles in quadrilateral is 360° , so $\angle AOB = 180^\circ - 80^\circ = 100^\circ$.

3. In $\triangle OAP$ and $\triangle OBP$, by RHS congruence, $\angle POA = \angle POB$.
4. Therefore, $\angle POA = \frac{1}{2}\angle AOB = 50^\circ$.

Derivation:

$$\angle AOB = 180^\circ - 80^\circ = 100^\circ; \angle POA = \frac{100^\circ}{2} = 50^\circ$$

Circles — Tangent properties

- 19. 19.** Prove that the parallelogram circumscribing a circle is a rhombus.

Answer:

$$AB = BC = CD = DA$$

Solution:

1. Since it is a parallelogram, opposite sides are equal: $AB = CD$ and $AD = BC$.
2. From the property of circumscribed quadrilaterals, $AB + CD = AD + BC$.
3. Substitute CD with AB and BC with AD : $2AB = 2AD$, hence $AB = AD$.
4. Since adjacent sides are equal, the parallelogram is a rhombus.

Derivation:

$$\begin{aligned} AB + CD &= AD + BC \\ \implies 2AB &= 2AD \\ \implies AB &= AD \end{aligned}$$

Circles — Tangents to a circle

- 20. 20.** A point P is 13 cm from the centre of a circle. The length of the tangent drawn from P to the circle is 12 cm. Find the radius of the circle.

Answer:

$$5 \text{ cm}$$

Solution:

1. Let O be the center and A be the point of contact on the circle.
2. Triangle OAP is a right-angled triangle at A because the radius is perpendicular to the tangent.
3. *Hypotenuse* $OP = 13\text{cm}$, *tangent* $AP = 12\text{cm}$.
4. Using Pythagoras theorem, $OA^2 = OP^2 - AP^2 = 13^2 - 12^2 = 169 - 144 = 25$.
4. $OA = 5\text{cm}$.

Derivation:

$$OA = \sqrt{13^2 - 12^2} = \sqrt{169 - 144} = \sqrt{25} = 5 \text{ cm}$$

Circles — Tangents to a circle

- 21. 21.** Prove that the lengths of tangents drawn from an external point to a circle are equal.

Answer:

$$PA = PB$$

Solution:

1. Draw a circle with center O and an external point P.
2. Draw tangents PA and PB touching the circle at A and B respectively.
3. Join OA, OB, and OP.
4. Consider triangles OAP and OBP.
5. Use the property that the tangent at any point of a circle is perpendicular to the radius through the point of contact.
6. Use RHS congruence criterion to prove triangles are congruent.
7. Conclude that corresponding parts of congruent triangles are equal.

Derivation:

In $\triangle OAP$ and $\triangle OBP$:

$$1. \angle OAP = \angle OBP = 90^\circ (\text{Tangent} \perp \text{radius})$$

$$2. OP = OP (\text{Common side})$$

$$3. OA = OB (\text{Radii of the same circle})$$

By RHS congruence rule, $\triangle OAP \cong \triangle OBP$.

Therefore, $PA = PB$ (CPCT).

Circles — Tangents to a circle

- 22. 22.** Two concentric circles are of radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle.

Answer:

$$8\text{ cm}$$

Solution:

1. Let O be the center of both circles.
2. Let AB be the chord of the larger circle touching the smaller circle at point P.
3. OP is the radius of the smaller circle, so $OP = 3$ cm.
4. OA is the radius of the larger circle, so $OA = 5$ cm.
5. Since AB is a tangent to the inner circle, $OP \perp AB$.
6. In right $\triangle OPA$, use Pythagoras theorem to find AP.
7. Since the perpendicular from the center bisects the chord, $AB = 2 \times AP$.

Derivation:

$$\text{In } \triangle OPA : OA^2 = OP^2 + AP^2$$

$$5^2 = 3^2 + AP^2$$

$$25 = 9 + AP^2$$

$$AP^2 = 16$$

$$\Rightarrow AP = 4 \text{ cm.}$$

Since $OP \perp AB$, $AP = PB = 4$ cm.

$$AB = AP + PB = 4 + 4 = 8 \text{ cm.}$$

- 23. 23.** A quadrilateral $ABCD$ is drawn to circumscribe a circle. Prove that $AB + CD = AD + BC$.

Answer:

$$AB + CD = AD + BC$$

Solution:

1. Identify the points of contact on sides AB , BC , CD , and DA as P , Q , R , and S respectively.
2. Use the theorem that lengths of tangents from an external point to a circle are equal.
3. Express segments AP , BP , BQ , CQ , CR , DR , DS , and AS in terms of equality.
4. Sum the equations formed by these segments.
5. Rearrange the terms to group segments forming the sides of the quadrilateral.

Derivation:

$$AP = AS(1), BP = BQ(2), CR = CQ(3), DR = DS(4)$$

$$\text{Add}(1), (2), (3), \text{and}(4) :$$

$$(AP + BP) + (CR + DR) = (AS + DS) + (BQ + CQ)$$

$$AB + CD = AD + BC.$$

Circles — Tangents to a circle

- 24. 24.** If a hexagon $ABCDEF$ circumscribes a circle, prove that $AB + CD + EF = BC + DE + FA$.

Answer:

$$AB + CD + EF = BC + DE + FA$$

Solution:

1. Let the points of contact be P , Q , R , S , T , and U on sides AB , BC , CD , DE , EF , and FA respectively.
2. By the theorem of tangents from an external point, $AP = AF$, $BP = BQ$, $CQ = CR$, $DR = DS$, $ES = ET$, $FT = FU$ is incorrect; rather, label them appropriately.
3. Correct logic: $AP = AU$, $BP = BQ$, $CQ = CR$, $DR = DS$, $ES = ET$, $FT = FU$.
4. Sum the left side $AB + CD + EF$ using these relations.
5. Observe that the sum equals the sum of the remaining sides $BC + DE + FA$.

Derivation:

Let points of contact be $P_1, P_2,$

$$AB + CD + EF = (AP_1 + P_1B) + (CP_3 + P_3D) + (EP_5 + P_5F) = (AP_6 + P_2B) + (CP_2 + P_4D) +$$

$$\text{Rearranging : } (P_2B + CP_2) + (P_4D + EP_4) + (P_6F + AP_6) = BC +$$

Circles — Tangents to a circle